

Data Analytics in Embedded Systems

- Uses **historical and real-time data**, Applies **rules, thresholds without learning framework**
- **Explains what happened and what is happening**
- Deterministic and Runs on microcontrollers (MCUs)
- **Embedded Use Case: Battery Management**

System (BMS):

- State of Charge (SoC) Algorithm
- Average temperature with Analytics techniques
- Power consumption trends with past and real time data

Result: Trigger alerts if thresholds exceeded

Evolution of Intelligence in Embedded Systems

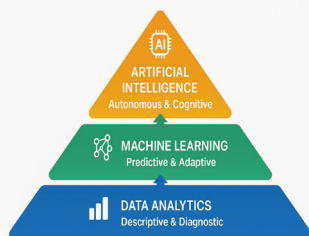


Fig. 1: Overview of the workflow of traditional AI principles

and integration testing. Alongside this, automotive development also requires strong compliance adherence through standards such as ISO 26262, ASPICE, and cybersecurity frameworks, where traceability, version control, testing, defect analysis, documentation, and risk assessment become critical but highly time-consuming activities.

The manufacturing and deployment workflow follows a similar structure. Once software updates or change requests are received from the production side, developers analyse their impact on requirements, architecture, design and testing before deploying updates to tier-one servers and production plants. Today, much of this process remains manual, including defect fixing, documentation, validation, and release management. This is where AI can play a major role by automating impact analysis, test generation, compliance checks, and workflow optimisation while still keeping human validation in the loop.

At the same time, automotive embedded systems are increasingly driven by three major compliance pillars: ASPICE, functional safety, and cybersecurity. Functional safety focuses on how safely and controllably a system operates under different conditions, while cybersecurity focuses on identifying and mitigating threats when ECUs are connected to networks or receive over-the-air updates. Processes such as HARA for hazard analysis and TARA for threat

analysis require different mindsets and extensive documentation. The key question is: How can AI help simplify these complex workflows, improve compliance management, and reduce overall development time?

AI vs ML vs data analytics: What really matters?

So far, we have examined the compliance requirements and the basic embedded system workflow. Now, we move on to the different principles of AI and how they have evolved over time.

AI did not suddenly appear overnight. Many of its foundations are decades old, but today we recognise and organise these concepts under the umbrella of artificial intelligence. AI is not fully autonomous and cannot operate without human involvement, as developers still create the rules, logic, and workflows behind it. In many ways, the intelligence behind AI still originates from the people designing the system.

Before AI, there was data analytics, which is more descriptive and diagnostic in nature. It focuses on understanding patterns, analysing trends, and interpreting what happened and what may happen next. In embedded systems, particularly in battery management systems (BMSs) used in EVs, data analytics plays a major role in monitoring battery charge levels, temperature behaviour, power consumption trends, and cell-balancing strategies. Developers can analyse

battery degradation and determine whether active or passive charging methods are more suitable using collected sensor data. These systems are deterministic and lightweight, allowing them to run efficiently on microcontrollers without adding significant complexity.

Next comes machine learning (ML). This is where the focus shifts from analysis to prediction. Machine learning models require labelled datasets and training layers to classify patterns and predict outcomes. This differs from traditional data analytics. ML is particularly useful in embedded systems for applications such as motor fault detection, where algorithms can identify abnormal behaviour and predict motor degradation or failure during runtime.

In layman's terms:

- Data analytics tells us what is happening
- Machine learning predicts what may happen
- Artificial intelligence takes action based on those predictions

Taking this a step further, artificial intelligence combines analytics, ML models, optimisation techniques, and automated decision-making. However, deploying AI in embedded systems is not straightforward because it depends heavily on hardware compatibility, microcontroller capabilities, compiler support, and dedicated accelerators such as NPUs. Modern microcontrollers and SoCs are now being designed with neural processing units (NPUs) to efficiently support AI workloads in constrained embedded environments.

One of the most critical use cases for embedded AI today is ADAS (advanced driver assistance systems). In these applications, systems must react within fractions of a second because outcomes directly affect the safety of drivers and pedestrians. Unlike analytics, which only studies data, or ML, which predicts outcomes, AI can actively respond to situations.